Op-amps, differential amps, power amps

Practice questions · Day 21 · Fri 13 Nov · pack E13

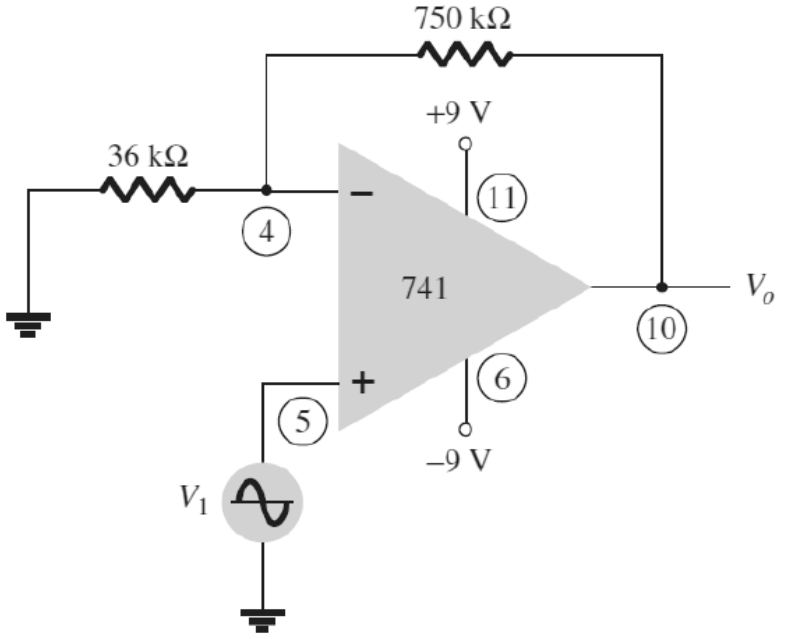
5. Practice

From your course files

WS10 Q1 (course question). Calculate the CMRR (in dB) for the circuit measurements of $V_d = 1$ mV, $V_o = 120$ mV, $V_C = 1$ mV, and $V_o = 20$ mV.

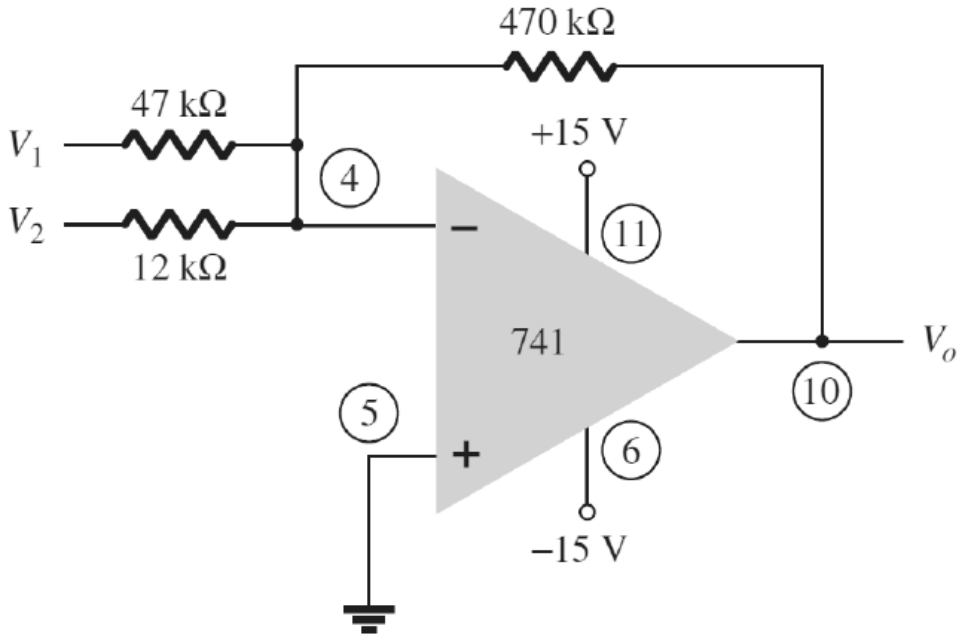
WS10 Q2 (course question). Determine the output voltage of an op-amp for input voltages of $V_{i1} = 200$ mV and $V_{i2} = 140$ mV. The amplifier has a differential gain of $A_d = 6000$ and the value of CMRR is: (a) 200 and (b) 10^5.

WS10 Q3 (course question). Calculate the output voltage of the circuit of Fig. 1 for an input of 150 mV rms.



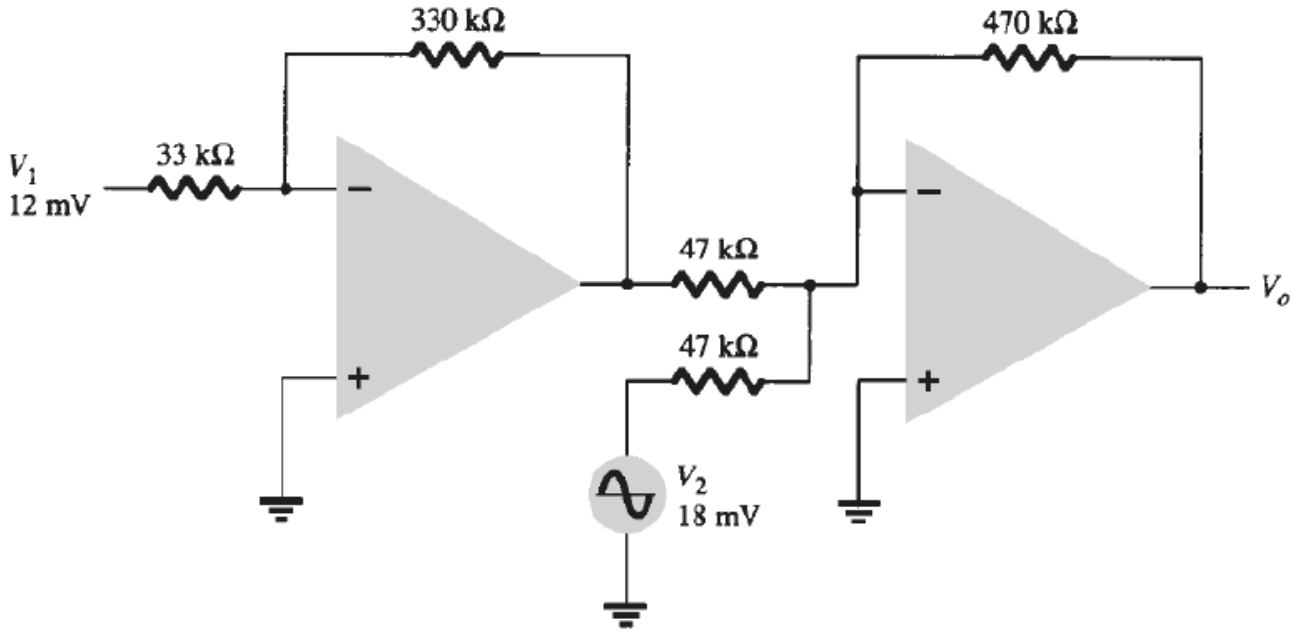
WS10 Fig. 1, page 2 (from the workshop sheet)

WS10 Q4 (course question). Calculate the output voltage for the circuit of Fig. 2 with inputs of $V_1 = 40$ mV rms and $V_2 = 20$ mV rms.



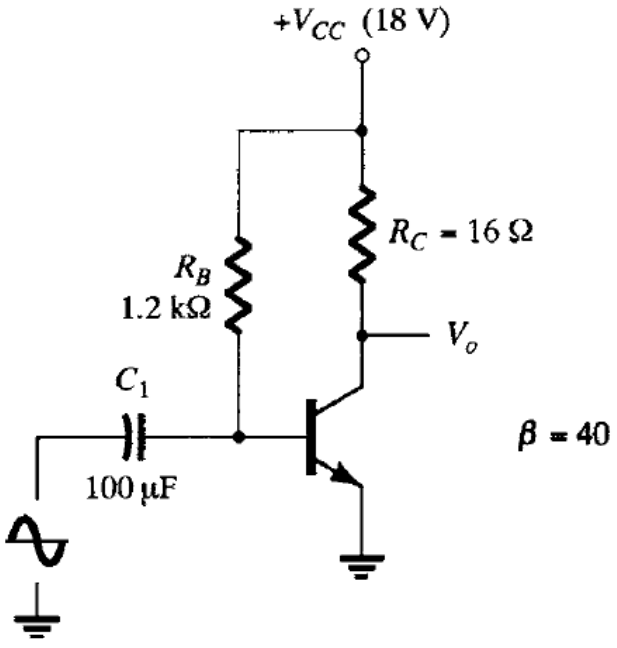
WS10 Fig. 2 (from the workshop sheet)

WS10 Q5 (course question). Determine the output voltage for the circuit of Fig. 3.



WS10 Fig. 3 (from the workshop sheet)

WS10 Q6 (course question). Calculate the input and output power for the circuit of Fig. 1. The input signal results in a base current of 5 mA rms.



WS10 Fig. 1, page 3 (from the workshop sheet)

WS10 Q7 (course question). Calculate the input power dissipated by the circuit of Fig. 1 if R_B is changed to 1.5 kΩ. *(Fig. 1 here is the class A circuit shown under WS10 Q6.)*

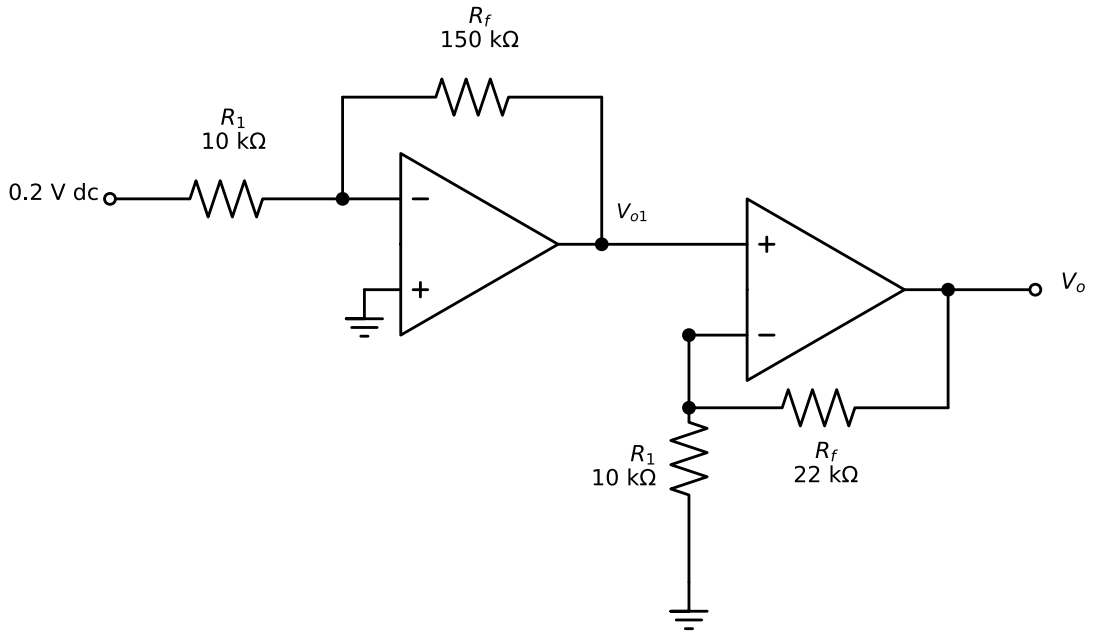
WS10 Q8 (course question). What maximum output power can be delivered by the circuit of Fig. 1 if R_B is changed to 1.5 kΩ? *(Same class A circuit as WS10 Q6.)*

WS10 Q9 (course question). Calculate the transformer turns ratio required to connect four parallel 16-Ω speakers so that they appear as an 8-kΩ effective load.

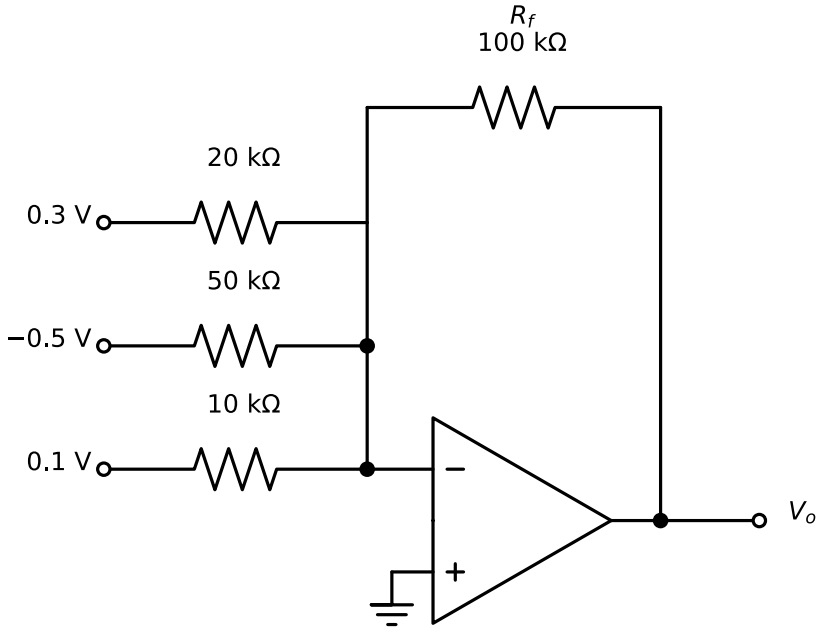
WS10 Q10 (course question). For a class B amplifier with $V_{CC} = 25$ V driving an 8-Ω load, determine: (a) Maximum input power, (b) Maximum output power, and (c) Maximum circuit efficiency.

New questions

N1. [3 marks] An op-amp has $A_d = 5000$ and CMRR = 80 dB. (a) Find A_c. (b) Find V_o for $V_{i1} = 1.25$ mV and $V_{i2} = 0.75$ mV.

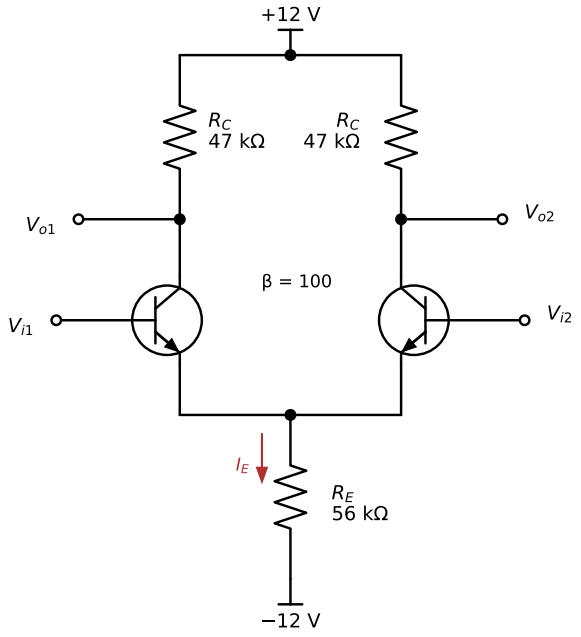


Circuit for N2(a): inverting stage into a non-inverting stage



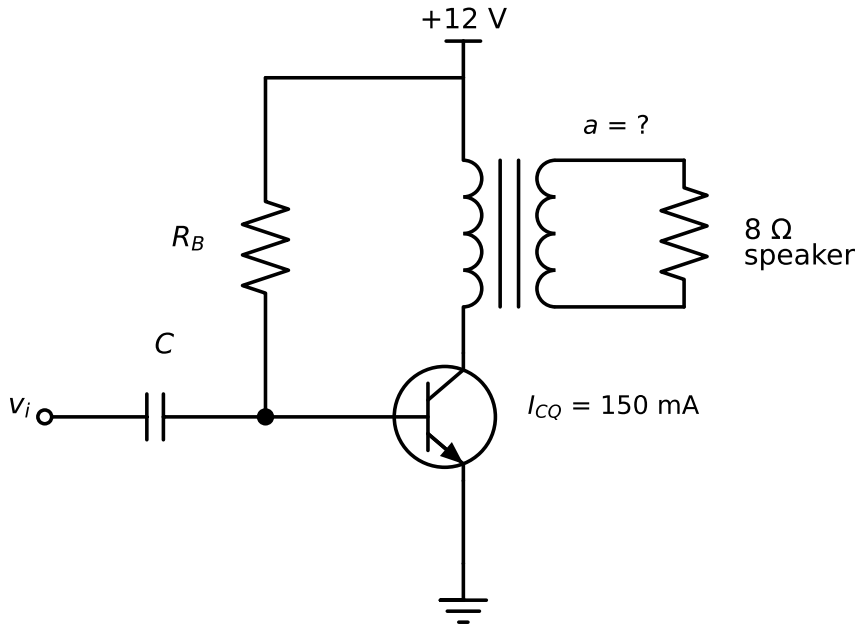
Circuit for N2(b): three-input inverting summer

N2. [5 marks] (a) A 0.2 V dc input drives an inverting stage with $R_1 = 10$ kΩ and $R_f = 150$ kΩ. That output drives a non-inverting stage with $R_1 = 10$ kΩ and $R_f = 22$ kΩ. Find both stage outputs. (b) An inverting summer has $R_f = 100$ kΩ and inputs $V_1 = 0.3$ V through 20 kΩ, $V_2 = -0.5$ V through 50 kΩ, and $V_3 = 0.1$ V through 10 kΩ. Find V_o.



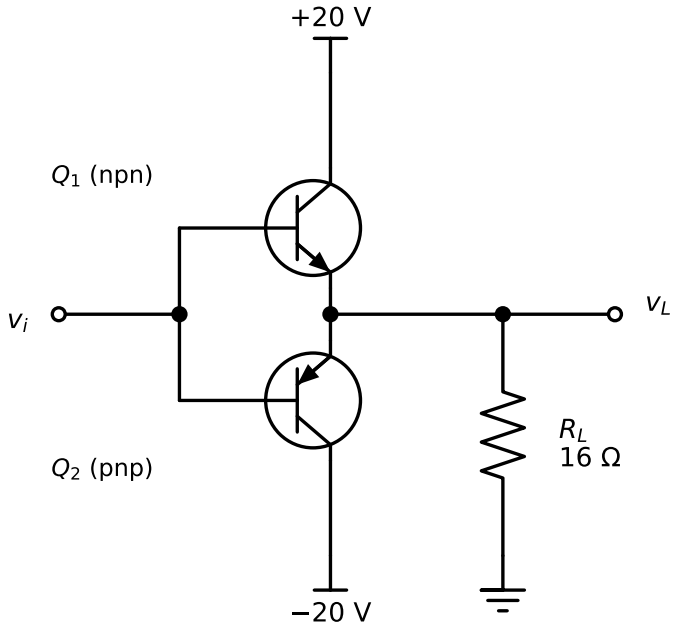
Circuit for N3: BJT differential amplifier

N3. [6 marks] A BJT differential amplifier has $V_{CC} = V_{EE} = 12$ V, tail resistor $R_E = 56$ kΩ, $R_C = 47$ kΩ on each collector, $\beta = 100$, and both bases at 0 V dc. Find (a) I_E (tail), I_C per transistor and V_C; (b) r_e; (c) the single-ended gain with one input driven, and the double-ended differential gain.



Circuit for N4: transformer-coupled class A amplifier

N4. [6 marks] A transformer-coupled class A amplifier runs from $V_{CC} = 12$ V with $I_{CQ} = 150$ mA. The signal swings V_{CE} between 2 V and 22 V and I_C between 10 mA and 290 mA. The speaker is 8 Ω. Find (a) P_o, (b) P_i and η, (c) the transistor dissipation, (d) the effective load seen by the collector and the turns ratio.



Circuit for N5: class B push-pull, V_CC = 20 V, R_L = 16 ohm

N5. [8 marks, chained] A class B push-pull amplifier uses $V_{CC} = 20$ V into $R_L = 16\ \Omega$. (a) For a peak output of 16 V, find P_i, P_o, η, and the dissipation per transistor. (b) Find the maximum possible P_o. (c) Find the worst-case total transistor dissipation and the output peak voltage at which it occurs. (d) Explain in one sentence why the transistors need heatsinks sized for (c), not for (b).

6. Quick recall check

1. $V_{i1} = 3$ mV, $V_{i2} = 1$ mV. What are V_d and V_c?
2. CMRR = 100 dB as a ratio?
3. Gain of a non-inverting amp with $R_f = 90$ kΩ, $R_1 = 10$ kΩ?
4. Maximum efficiency of series-fed class A, transformer-coupled class A and class B?
5. A 10 Ω load must look like 1 kΩ. What is the turns ratio?